AdaDINO-LA: A Controlled Study of DINOv2 Adaptation Strategies for Left Atrial Segmentation in LGE-MRI

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Abstract

Left atrial (LA) segmentation from late gadolinium-enhanced MRI (LGE-MRI) supports atrial fibrillation studies, but manual delineation is slow and observer-dependent. Task-specific networks such as nnU-Net, the established reference, are trained from scratch and need annotations for each dataset. Pretrained foundation encoders such as DINOv2 may reduce this need. Such encoders can be frozen, partially or fully fine-tuned, or adapted with low-rank adaptation (LoRA), but to our knowledge no previous LA study compared them under one protocol. We compared the four strategies on DINOv2 ViT-Large and three on ViT-Giant, with the same decoder, loss, model-selection rule, three seeds, and 154-volume split (by volume; patient identifiers were unavailable). Pretraining mattered most: the same ViT reached a Dice coefficient of 0.764 from scratch and 0.907 from DINOv2. Adaptation strategy changed Dice by only 1 to 3 percentage points; ViT-Giant, at about three times the memory, and a skip-connection decoder each added at most 0.6. With one improved recipe applied post hoc to all ViT-Large strategies, LoRA reached the mean Dice of full fine-tuning (0.912) with 56% of its memory. With this recipe and 11 of 108 training volumes, the pretraining gain grew from 15 to 42 points. The nnU-Net reference, which alone used augmentation, exceeded the best DINOv2 model internally by 1.6 points (3.7 with 11 volumes) and on bSSFP MRI, but not clearly on external LGE data. DINOv2 thus did not surpass nnU-Net, but within its pipeline a well-tuned recipe mattered more than a larger backbone, with LoRA as a lower-memory option.

Keywords: Left atrial segmentation, LGE-MRI, DINOv2, Fine-tuning, Low-rank adaptation, Computational efficiency

1. Introduction

Accurate delineation of the left atrium (LA) supports quantitative analysis of cardiac magnetic resonance images in atrial fibrillation studies, including scans acquired before and after catheter ablation [1]. In late gadolinium-enhanced magnetic resonance imaging (LGE-MRI), atrial anatomy must be distinguished from surrounding tissue despite variable image contrast and complex boundaries. Manual annotation is labor-intensive and subject to intra- and inter-observer variability, motivating automated segmentation methods that produce consistent anatomical masks [1, 2]. The present study addresses the binary LA endocardial segmentation target supplied with the dataset, rather than atrial scar classification or clinical outcome prediction.

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Task-specific frameworks such as nnU-Net, configured and trained from scratch for each dataset, remain strong references for medical segmentation [3–5]. Because these networks learn all of their features from the annotated scans of the target dataset, each new dataset or acquisition setting typically requires its own labeled training set. Self-supervised vision foundation models, notably DINOv2, offer a different route: patch-level features are learned once from large unlabeled natural-image collections and then reused for dense prediction [6]. Reusing pretrained features may reduce the number of annotated scans needed for training [7] and improve robustness to differences between scanners and acquisition protocols [8]. For LA segmentation, DINOv2 encoders have been reported to remain accurate with limited training data [9] and, combined with U-Net-style decoders, to exceed nnU-Net [10]. A practitioner adopting a DINOv2 encoder must decide whether to keep it frozen, partially fine-tune it, adapt it with low-rank adaptation (LoRA) [11], or fully fine-tune it; whether a larger backbone is worth its cost; and how much of the resulting accuracy is owed to pretraining, particularly when few scans are annotated.

Existing evidence does not resolve these decisions for LA segmentation. The closest studies, by Kundu et al., evaluated DINOv2 across backbone sizes [9], paired it with different decoders [10], and added multi-scale feature fusion [12]. Because the decoder, data partition, and reference pipeline differed between these studies, differences in their results cannot be attributed to a single factor. For medical segmentation more broadly, benchmarks have compared fine-tuning strategies for SAM across modalities [13], parameter-efficient adaptation for few-shot volumetric organ segmentation [14], and transfer paradigms for several foundation models on 3D tasks [15]. None focused on LA segmentation in LGE-MRI, and in one the advantage of parameter-efficient over full fine-tuning depended on backbone size [13], so their rankings may not carry over to this task. The contribution of pretraining cannot be assumed either, since natural-image pretraining does not always benefit medical tasks [16]. Resolving these decisions requires one protocol in which adaptation strategy, backbone scale, and pretraining are varied while other factors are held fixed, because rankings of segmentation methods can change with evaluation choices [17]. Training resources must also be measured directly, because the number of trainable parameters does not by itself determine training memory or time.

We therefore conduct a controlled empirical comparison, termed AdaDINO-LA, in which the principal experimental design factors are held fixed where possible. AdaDINO-LA denotes the study design, not a new network, and control here means a shared protocol with nominally fixed hyperparameters rather than per-strategy tuning. The study addresses four research questions:

RQ1. How much does DINOv2 initialization contribute compared with the same architecture trained from random initialization, and how do this contribution and the gap to a task-specific nnU-Net change when fewer annotated training volumes are available?

RQ2. Under the shared protocol, how do adaptation strategy and backbone scale affect accuracy and training resources?

RQ3. How sensitive are these effects to the training recipe, namely the adapter learning rate, gradient clipping, the decoder readout, and decoder capacity, and how do the strategies compare when all of them receive one common improved recipe?

RQ4. How do the adapted models perform on external data from other centers and from another MRI sequence, compared with a task-specific nnU-Net?

The study makes three empirical contributions. First, it compares four adaptation strategies and two backbone sizes under one split, preprocessing recipe, decoder topology, loss implementation, and checkpoint rule, with three seeds, paired case-level statistics, and a no-pretraining control. Second, it measures training memory, throughput, and estimated training cost. Third, it tests which conclusions persist under recipe changes, including one common improved recipe applied

to all ViT-Large strategies; with a larger decoder with skip connections; under reduced label budgets; and on external data, with nnU-Net as a reference on the same held-out cases. Some findings, such as the gap between parameter and compute efficiency, confirm general experience; here they are quantified for LA segmentation under one protocol and condensed into practical recommendations.

2. Related work

2.1. Left atrial segmentation and task-specific pipelines

U-Net introduced a widely adopted encoder–decoder design for biomedical segmentation, and nnU-Net showed the importance of configuring preprocessing, architecture, and training together [3, 4]; its revisited evaluation emphasized rigorous validation against strong baselines [5]. UNETR connects a transformer encoder to a convolutional decoder through multi-resolution skip connections for volumetric segmentation [18]. Left atrial analysis from LGE-MRI includes cavity, wall, and scar segmentation, which are not interchangeable targets [1]. Usformer develops a compact network for 3D LA segmentation [19], and CA-Net combines Transformer and V-Net components in a semi-supervised framework [20]. These studies develop task-specific architectures; AdaDINO-LA instead compares adaptation configurations of one pretrained encoder within a common decoder topology and evaluation recipe. Scores reported on other datasets, partitions, or targets are not treated as comparable benchmarks.

2.2. Vision foundation models in medical imaging

DINOv2 provides patch tokens that can be reshaped into spatial feature maps [6], which makes frozen extraction a natural baseline for dense prediction. DINOv2 has been evaluated on radiology benchmarks [7] and, together with other foundation models, for domain generalization in medical segmentation [8]. Self-supervised pretraining has also been studied directly in 3D medical imaging, for example a Swin UNETR encoder pretrained on CT volumes [21], a medical-domain setting that differs from transferring natural-image representations.

Kundu et al. evaluated DINOv2 for LA segmentation across backbone sizes and limited-data settings on LAScarQS 2022 [9]. A second study paired DINOv2, SAM, and MedSAM encoders with linear or U-Net-style decoders on LAScarQS 2022 and on the 2018 challenge data used here; with the U-Net-style decoder, DINOv2 exceeded nnU-Net on both, whereas with linear decoders nnU-Net was better on the 2018 data [10]. A third used a frozen DINOv2 encoder with learned multi-scale fusion and image information in the decoding pathway and reported higher Dice than a 2D nnU-Net baseline whose configuration was not specified; repeated training seeds were not reported [12]. These results are revisited in Section 6.2.

Beyond DINOv2, MedSAM adapts a promptable segmentation model across modalities [22], a large-scale evaluation documents the limits of direct SAM transfer to medical images [23], and specialization and one-shot studies compare foundation-model pipelines with nnU-Net [24, 25]. These studies motivate evaluating transfer against strong task-specific references; their supervision regimes and targets differ from ours.

2.3. Parameter-efficient adaptation

LoRA represents a weight update as the product of two low-rank matrices while retaining the pretrained weights [11]; partial fine-tuning instead updates a subset of the original layers. Parameter-efficient adaptation was first benchmarked in medical imaging for classification and generation [26]. In medical segmentation, SAMed trains LoRA updates in the SAM image encoder together with the mask decoder [27], Medical SAM Adapter and Trans-SAM insert

adapter modules [28, 29], HF-SAM combines LoRA with multi-scale fusion for few-shot segmentation [30], and LoRA has been combined with a DeepLabV3 decoder to adapt DINOv2 for oral lesion segmentation [31]. Parameter efficiency and computational efficiency are distinct: frozen weights still participate in the forward pass, and adapters placed throughout a backbone require gradients to propagate through it. A small trainable fraction therefore need not imply a small memory footprint or short training time; a SAM benchmark found that parameter-efficient strategies improved accuracy without improving training efficiency [13]. AdaDINO-LA does not introduce a new adapter; it compares frozen extraction, partial fine-tuning, LoRA, and full fine-tuning within DINOv2 and reports several resource measures. Table 1 summarizes these distinctions.

Table 1. Positioning relative to selected prior work. Entries describe research objectives rather than an exhaustive absence/presence audit. Published scores are not compared across protocols.

Study	Main setting	Focus relative to this study
Usformer; CA-Net [19, 20]	Task-specific 3D LA segmentation	Architectural development; CA-Net uses semi-supervised learning.
Kundu et al. [9]	DINOv2 for LA segmentation	Backbone-size and limited-data evaluation on LAScarQS; directly related transfer setting.
Kundu et al. [10]	DINOv2, SAM, and MedSAM encoders for LA segmentation	Linear versus U-Net-style decoders on LAScarQS and the 2018 challenge data.
Kundu et al. [12]	LA segmentation with multi-scale fusion	Learned fusion and image-driven spatial integration in the prediction pathway.
SAMed; medical adapters; HF-SAM [27–30]	SAM-derived medical segmentation	LoRA/adapters and task-specific decoding or fusion mechanisms.
Adaptation benchmarks [13–15]	Fine-tuning strategies for foundation models across organs and modalities	Strategy rankings across backbones and decoders; not LA LGE-MRI.
AdaDINO-LA	DINOv2 for LGE-MRI LA segmentation	Common decoder topology and evaluation; four adaptation strategies and two backbone sizes over three seeds; no-pretraining control; recipe sensitivity; external evaluation; multiple resource measures.

3. Method

3.1. Problem formulation and study design

Let X_i denote an LGE-MRI volume and Y_i its binary reference mask. For slice z, preprocessing produces an input $x_{i,z} \in \mathbb{R}^{3\times448\times448}$. An encoder f_θ and decoder g_ϕ predict logits

$$s_{i,z} = g_\phi(f_\theta(x_{i,z})), \qquad p_{i,z} = \mathrm{sigmoid}(s_{i,z}). \tag{1}$$

Thresholded masks are restored to the native in-plane geometry and stacked to obtain the volume prediction $\widehat{Y}_i$. The network processes 2D slices; the evaluation unit is a reconstructed 3D volume.

On ViT-Large we compare frozen extraction, partial fine-tuning, LoRA, and full fine-tuning; on ViT-Giant we compare frozen extraction, LoRA, and partial fine-tuning of the last four blocks. Giant Full FT was not run because, at batch size 4, it was not expected to fit the

24 GB GPUs used for most runs (estimated from the 15.9 GB Giant LoRA peak plus about 14 GB of gradients and optimizer state). These shared strategies support matched backbone comparisons. Configuration names combine backbone and strategy, abbreviating fine-tuning as FT (for example, Large Full FT for full fine-tuning of ViT-Large). A Large frozen configuration trained with binary cross-entropy (BCE) alone evaluates the loss choice; all other configurations use the combined objective below. The identifiers E0–E7 used in the supplementary material link results to archived runs and are listed there.

3.2. DINOv2-based segmentation framework

We use pretrained DINOv2 ViT-Large/14 and ViT-Giant/14 encoders [6], with embedding dimensions $C = 1024$ and $C = 1536$. Both use 14×14 patches and therefore produce the same 32×32 grid of $N_{\text{tok}} = 1024$ spatial tokens at the selected resolution, so the decoder topology is unchanged; replacing one backbone with the other is not an isolated change in parameter count. Figure 1 summarizes the pipeline. Normalized patch tokens from the final encoder block are reshaped to $C \times 32 \times 32$, and non-spatial tokens are omitted. A 1×1 convolution projects the features to 256 channels, followed by batch normalization and ReLU. Three bilinear upsampling blocks (3×3 convolution, batch normalization, ReLU; 128, 64, and 32 channels) raise the resolution to 256×256. A final interpolation to 448×448 is followed by a 16-channel convolutional block and a one-channel output. The decoder is structurally common but its input projection depends on C; it has no encoder skip connections.

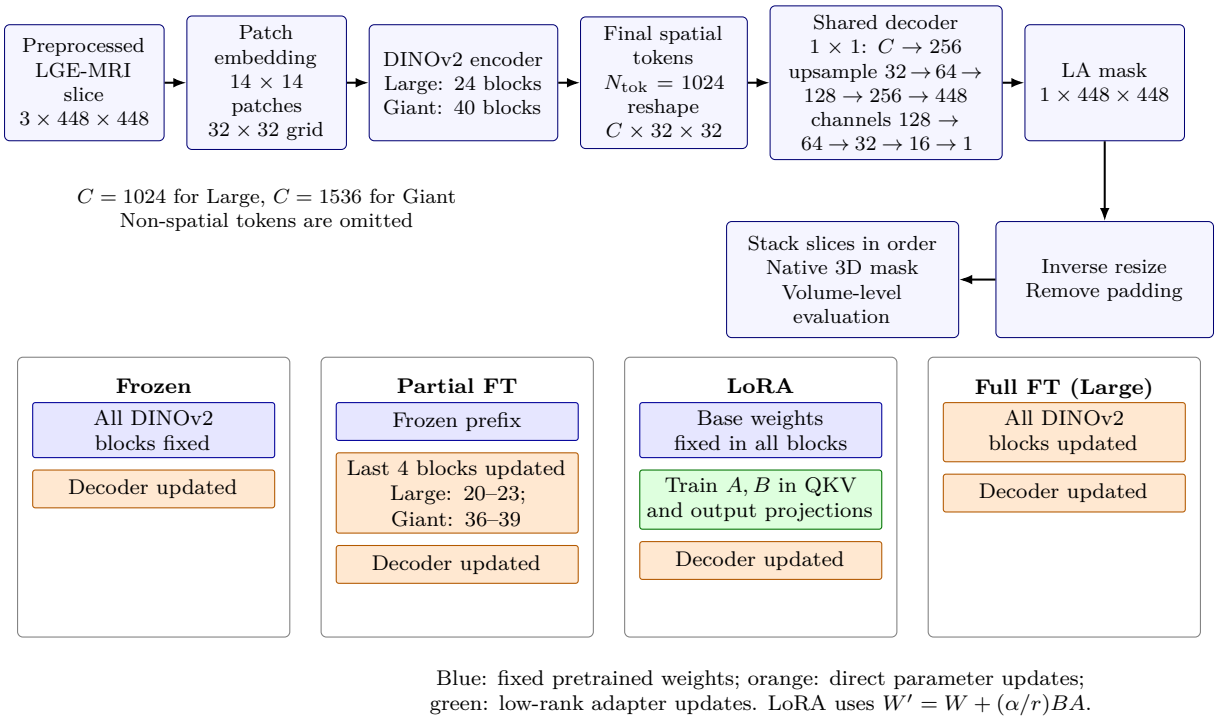


Figure 1. DINOv2 segmentation framework and adaptation mechanisms. The upper panel shows the patch embedding, encoder depth, token grid, backbone-dependent feature dimension, and shared decoder topology; the lower panel identifies which components are fixed, directly updated, or updated through LoRA. All four strategies are evaluated within Large; Giant covers Frozen, LoRA, and Partial FT. In the four-layer readout of the recipe analyses (Section 4.4), tokens from blocks 5, 11, 17, and 23 replace the final-block tokens and are projected separately. Stacking slice predictions enables native-volume evaluation but does not introduce 3D contextual learning.

3.3. Adaptation strategies

Frozen. All encoder parameters are frozen and the encoder runs in evaluation mode; only the decoder is optimized.

Partial fine-tuning. The last four Transformer blocks (zero-based indices 20–23 for Large and 36–39 for Giant) are updated together with the decoder; the patch embedding, earlier blocks, and final normalization remain frozen. Four blocks are 4/24 of Large depth but 4/40 of Giant depth, so the comparison matches block count, not depth fraction or trainable parameter count.

Full fine-tuning. All ViT-Large encoder parameters and the decoder are trainable, with a smaller learning rate for the encoder than for the newly initialized decoder.

LoRA. For a pretrained linear weight $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, the effective weight is

$$W' = W + \frac{\alpha}{r} BA, \quad A \in \mathbb{R}^{r \times d_{\text{in}}}, \quad B \in \mathbb{R}^{d_{\text{out}} \times r}, \tag{2}$$

with W frozen. Adapters are injected into the joint query–key–value projection and the attention output projection of every block; A uses Kaiming uniform initialization and B zeros, so the initial update is zero. Both backbones use the same rank and scaling (Table 2).

3.4. Training objective

For a batch of B slices with N pixels each, we optimize

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{BN} \sum_{k=1}^{B} \sum_{j=1}^{N} \left[y_{kj} \log p_{kj} + (1 - y_{kj}) \log(1 - p_{kj}) \right], \tag{3}$$

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{1}{B} \sum_{k=1}^{B} \frac{2 \sum_j p_{kj} y_{kj} + \epsilon}{\sum_j p_{kj} + \sum_j y_{kj} + \epsilon}, \tag{4}$$

$$\mathcal{L} = \tfrac{1}{2} \mathcal{L}_{\text{BCE}} + \tfrac{1}{2} \mathcal{L}_{\text{Dice}}, \qquad \epsilon = 10^{-6}. \tag{5}$$

BCE is computed from logits, and soft Dice is computed per slice and averaged over the batch. The BCE-only control changes the BCE weight from one half to one as well as removing Dice, so it measures objective sensitivity for frozen features rather than the effect of adding Dice alone. No claim is made that the shared objective is optimal for each strategy.

4. Experimental setup

4.1. Dataset and fixed case-level split

We study 154 LGE-MRI volumes of the local Utah2018 collection, each with an LA endocardial reference mask; the benchmark is described by Xiong et al. [2]. Our partition is not the official benchmark split. Every configuration uses the same 108 training, 15 validation, and 31 test volumes (split seed 2024). All 88 slices of each volume are retained, including slices without LA, giving 9,504, 1,320, and 2,728 slices.

A case denotes a volume rather than a verified patient, because the available identifiers do not establish a reliable volume-to-patient mapping. The collection includes pre- and post-ablation scans [32]. We therefore do not claim patient independence across subsets. An image-similarity audit of all 11,781 volume pairs, using array hashes, screening on a common grid, and native-resolution checks, identified one train–test pair with identical binary masks and highly correlated intensities (supplementary material). The audit detects near-duplicates, not patient identities. Excluding the identified test volume is reported as a sensitivity analysis.

4.2. Preprocessing and training

Intensities are clipped at the 0.5th and 99.5th percentiles of each volume (finite nonzero values when zero background is present), min–max normalized, and stored as 8-bit slices. Each slice is zero-padded to a square and resized to 448×448 (bilinear for images, nearest-neighbor for masks), replicated across three channels, and normalized with ImageNet statistics. No augmentation is used, so the comparisons concern adaptation under a fixed input distribution; this defines the protocol rather than asserting that augmentation would not help.

Each DINOv2 configuration is trained with seeds 42, 2026, and 3407 under the settings of Table 2. Decoder parameters are newly initialized for each run, and trainable-parameter counts and gradient checks confirm that the intended groups are updated. Forward computation uses mixed precision with dynamic gradient scaling, and every configuration computes the loss in FP32. The first runs of four configurations computed the loss under autocast; these were retrained for all seeds and only the retrained runs are reported. At each epoch, validation predictions are reconstructed at native resolution, and the checkpoint with the highest mean 3D Dice over the 15 validation volumes is kept; training stops after ten epochs without improvement. Test metrics are never used for selection.

Experiments ran on shared institutional GPUs (RTX 4090, RTX 5090, RTX PRO 6000, and RTX 4000 Ada; PyTorch 2.8.0 or 2.11.0; Table S8). Inference of the same checkpoint on two GPU models gave Dice identical to four decimals, whereas retraining the same configuration and seed on another GPU model changed Dice by 0.38 to 0.60 points in three pairs, of the order of seed-to-seed variation, so fixed seeds describe repeatability rather than bitwise determinism. The main Large Full FT runs also differ from the other main-table rows in GPU, PyTorch version, and DINOv2 source snapshot. Large language model coding assistants (Codex, OpenAI; Claude Code, Anthropic; used in 2026) helped write the data-preparation, experiment-management, evaluation, and analysis scripts; they were not used to create or modify images or labels. The authors reviewed all code, unit tests check the data preparation, model configurations, and evaluation, and the main-text result tables are regenerated from the released per-case results by the released scripts and checked to be identical to those in the manuscript.

4.3. Task-specific 3D reference

We train nnU-Net ResEncM 3D full-resolution (nnunetv2 2.8.1) as a task-specific reference [4, 5] with the same case lists, labels, and three seeds: it trains on the 108 training cases, selects its checkpoint on the 15 validation cases, and plans on these 123 development cases only. The baseline keeps its own recipe: Z-score normalization, $32 \times 256 \times 256$ patches, batch size two, nnU-Net augmentation, Dice plus cross-entropy with deep supervision, SGD with Nesterov momentum and polynomial decay, 1,000 epochs of 250 patch batches, checkpoint selection by a moving average of validation-patch Dice, and sliding-window inference with mirroring. Evaluation uses the same 31 held-out cases and evaluator. Its seeds fix only the main-process random generators, but the three runs produced distinct predictions. As a slice-wise counterpart, fixed post hoc together with the decoder study (Section 4.4), the ResEncM 2D configuration was trained with the same plans file, recipe, checkpoint rule, and seeds. Because dimensionality, normalization, augmentation, objective, optimization, budget, and inference differ jointly, the comparison is pipeline-level and does not isolate architecture or pretraining.

4.4. Pretraining control, recipe sensitivity, label budget, decoder capacity, and external evaluation

No-pretraining control (RQ1). The encoder and decoder of Large Full FT are trained with the identical protocol and seeds from random initialization, using DINOv2's own initialization scheme. Because the shared encoder learning rate is low for training from scratch, a second

Table 2. DINOv2 training settings of the main protocol. Backbone learning rate applies only to unfrozen parameters or LoRA adapters. Settings varied in the recipe-sensitivity analyses are described in Section 4.4.

Setting	Value
Pretrained encoders	DINOv2 `dinov2_vitl14` and `dinov2_vitg14` (non-register)
Input / patch / grid	448×448 / 14×14 / 32×32
Training seeds	42, 2026, 3407
Optimizer	AdamW; $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$
Learning rates	Decoder: 10^{-3}; backbone/adapters: 10^{-5}; constant schedule
Weight decay	0.01 for weight matrices; 0 for biases, normalization, and other one-dimensional parameters
Batch size	4 slices; gradient accumulation: 1
Partial FT blocks (zero-based)	Large: 20–23; Giant: 36–39; decoder trainable
Training limit	50 epochs; early-stopping patience: 10
Precision	AMP FP16; dynamic GradScaler; loss computed in FP32 for all configurations
Gradient clipping	None in the main protocol
LoRA configuration	$r = 4$, $\alpha = 4$; all attention qkv/proj layers; no LoRA dropout
Model selection	Highest mean validation volume Dice
Prediction	Sigmoid threshold 0.5; no postprocessing

recipe uses an encoder learning rate of 10^{-4} with gradient-norm clipping at 1.0, also with three seeds. Frozen, Partial, and LoRA variants are not evaluated without pretraining, because their frozen weights would be random.

Recipe sensitivity (RQ3). Each factor is first varied for the strategy it concerns, with three seeds. (i) Adapter learning rate: a single-seed validation sweep of Large LoRA selected 10^{-4} (10^{-3} diverged), which was then applied to Large and Giant LoRA. (ii) Gradient-norm clipping at 1.0 for Large Full FT. (iii) A four-layer decoder readout for Large Frozen and Large LoRA: tokens from blocks 5, 11, 17, and 23 are each projected to 64 channels and concatenated, which partitions the original $1024 \rightarrow 256$ projection without adding trainable parameters. (iv) A common improved protocol applies all three to the four ViT-Large strategies: the four-layer readout, clipping at 1.0, and a rule in which pretrained weights keep 10^{-5} and newly initialized parameters use a higher rate (decoder 10^{-3}, zero-initialized LoRA adapters 10^{-4}). Its 12 runs used one RTX 5090 and one software stack, and their test set was evaluated only after all of them had finished. All recipe analyses were defined after the main test results were known, select on validation data only, and leave the main-table configurations unchanged; several single-factor variants ran on different hardware from their reference runs (Table S8).

Label budget (RQ1). The common improved recipe (iv) is retrained with 11 and 27 of the 108 training volumes (10% and 25%). For each seed, one random permutation of the training volumes defines nested subsets, which all configurations of that seed share. The four ViT-Large strategies, nnU-Net, and the randomly initialized ViT-Large are trained on each subset with three seeds. The randomly initialized encoder uses the common recipe with full fine-tuning and an encoder learning rate of 10^{-4}, the rate given to newly initialized parameters; it is therefore not identical to either from-scratch recipe of the main analysis and is also trained on all 108 volumes, whereas the other 108-volume results are the common-recipe runs and the nnU-Net reference. A reduced-data epoch cycles through the subset until it has drawn the 2,376 batches of a full-data epoch, so the update schedule and early-stopping rule are unchanged; neither these nor any other setting were tuned per budget. Validation and test volumes are unchanged, so a

budget of 11 training volumes still uses 26 annotated development volumes, and nnU-Net keeps the plans derived from the 123 development volumes. Subsets and runs were fixed after the main and common-recipe test results were known but before any reduced-data run; a 50% budget was dropped for compute time before any run at that budget. All label-budget runs used one RTX 5090, and a rerun of one 108-volume LoRA run with this code reproduced its Dice within 0.11 points.

Decoder capacity (RQ3). The lightweight decoder of the common improved recipe is replaced by a hierarchical decoder in the style of UNETR [18]. Tokens from blocks 23, 17, 11, and 5 are each projected to 256 channels and enter a top-down path at 32, 64, 128, and 224 pixels, where they are concatenated with the upsampled path and fused by two convolutional blocks; a convolutional stem on the input slice adds a skip connection at 448 pixels. The decoder has 3.12 instead of 0.66 million parameters. The four ViT-Large strategies were retrained with it on all 108 training volumes, and LoRA and Full FT also on the 11-volume subsets, with the subsets, batch order, learning-rate rule, checkpoint rule, and evaluation of the common-recipe and label-budget runs, which serve as their lightweight-decoder counterparts. The 18 runs used one RTX 5090. Their design and planned contrasts (hierarchical minus lightweight decoder, nnU-Net minus hierarchical decoder, and the ranking of strategies) were fixed after the common-recipe and label-budget results were known but before any of these runs.

External evaluation (RQ4). Without retraining, all main-protocol checkpoints and nnU-Net are applied to two public datasets, and the common-recipe checkpoints are scored post hoc with the same protocol. CDEMRIS provides pre- and post-ablation LGE volumes of 30 patients [33]; the 10 patients whose volumes carry Utah-style headers (p6, p8, p13, p15, p16, p20, p22, p24, p27, p30) were excluded a priori to avoid overlap with the training data, leaving 40 volumes from 20 patients at two non-Utah centers. Task02 of the Medical Segmentation Decathlon (MSD), derived from the 2013 Left Atrial Segmentation Challenge, contains 30 volumes acquired with balanced steady-state free precession (bSSFP), a different MRI sequence [34, 35]; the 20 volumes with public reference masks are used, and their identity affines required anatomical orientation correction. Preprocessing, threshold, and native-grid evaluation follow the main protocol. Label definitions may differ between centers, so external Dice values are not compared with internal values.

4.5. Metrics and statistical analysis

Slice predictions are thresholded at 0.5, resized to the padded native geometry with nearest-neighbor interpolation, unpadded, and stacked. For binary volumes P and G, Dice $= 2|P \cap G|/(|P|+|G|)$ is the primary outcome and IoU $= |P \cap G|/|P \cup G|$ is complementary; both are one if both volumes are empty and zero if only one is. Metrics are computed per volume, averaged equally over test volumes for each seed, and summarized by the mean and sample standard deviation (SD) of the three seed means. The SD therefore describes retraining variation on one split. Dice differences are reported in percentage points (Dice $\times$ 100), abbreviated as points. The 95th-percentile Hausdorff distance is reported only as an exploratory supplementary summary because physical spacing has not been independently verified.

For paired comparisons, Dice is averaged over seeds within each case, and the 31 cases are resampled with replacement 20,000 times with all configurations kept paired (random seed 20260926, reset for each contrast). Percentile intervals give 95% confidence intervals (CIs), and two-sided p values are twice the smaller bootstrap tail probability; when no draw crosses zero, this resolution bounds Holm-adjusted values only by 0.002, which is reported as $p_{\mathrm{Holm}} < 0.002$. The 16 contrasts of Table 4, including three nnU-Net reference contrasts, are Holm-adjusted as one family, which was fixed after the main test results were known; the 15 label-budget contrasts of Table 7 form a second family, fixed after those results were viewed; and the 12 planned contrasts

of Table 8 form a third, fixed before those runs. Because each seed of the label-budget analysis also draws its own subset, its contrasts and those of the decoder study are additionally checked with a case × seed bootstrap that resamples the three seeds jointly across configurations. All tests are therefore exploratory: conclusions rest on effect sizes and intervals rather than on p values, and other contrasts carry unadjusted intervals. Because seeds are averaged first, the intervals are conditional on the trained models; per-seed differences are reported, and a contrast is called robust only when its interval excludes zero and all three per-seed differences share its sign, a minimum screen rather than a test. For CDEMRIS, the bootstrap resamples patients. Two robustness analyses re-estimate key contrasts (Table S4): a case × seed bootstrap whose intervals include retraining variation, and a restriction to the two thirds of test volumes least similar to any training volume in the image-similarity audit. The intervals assume exchangeable test volumes, which unverified patient independence may violate, and an interval containing zero does not establish equivalence.

Writing $D_{b,s}$ for mean Dice with backbone b and strategy s, we report adaptation effects $D_{b,s} - D_{b,\mathrm{Frozen}}$, scale effects $D_{\mathrm{Giant},s} - D_{\mathrm{Large},s}$, and scale interactions $I_s = \Delta_{\mathrm{scale}}(s) - \Delta_{\mathrm{scale}}(\mathrm{Frozen})$ for $s \in \{\mathrm{LoRA}, \mathrm{Partial}\}$, all formed within cases before resampling; changes between label budgets are formed in the same way. Training stability is summarized post hoc by the maximum drawdown of the validation trajectory, $\mathrm{MDD} = \max_{u \le t}(D_u - D_t)$, which captures declines even if the run later recovers.

4.6. Training resources

We report trainable parameters and peak allocated training memory, measured without gradient checkpointing. Throughput is measured with a fixed workload in which every configuration processes the same 300 batches on one RTX 4090 (supplementary material). Training cost is expressed as estimated RTX 4090 GPU-hours: measured time per batch multiplied by the training batches completed in each run, in total and up to the selected checkpoint. These estimates exclude validation and checkpoint writing and are labeled as estimates. A configuration is non-dominated for a resource if no other configuration has at least its mean Dice at no greater cost, with one strict improvement; frontiers are descriptive and are computed separately for memory, time, and trainable parameters. nnU-Net is excluded because its 3D patch workload is not matched to the 2D slice workload.

5. Results

5.1. Contribution of DINOv2 pretraining

Table 3 reports all configurations. The ViT-Large encoder and decoder trained from random initialization under the Large Full FT protocol reached Dice 0.7639 ± 0.0234. DINOv2 initialization was higher by 14.27 percentage points (95% CI [9.52, 20.38], unadjusted and outside the Holm family of Table 4; per-seed differences 14.23, 12.32, and 16.26), with higher Dice in all 31 test cases. Even the frozen DINOv2 encoder exceeded the randomly initialized encoder trained end to end, by 11.08 points ([6.36, 17.22]). The from-scratch recipe with a tenfold encoder learning rate and clipping was lower still, at 0.7322 ± 0.0084 (-3.17 points, $[-4.50, -1.98]$; two of its runs reached the 50-epoch limit), so the 14.3-point gap is measured against the stronger of the two from-scratch recipes. Under this protocol, the ViT encoder trained on 108 volumes therefore depended on DINOv2 initialization, and more so with fewer volumes (Section 5.5); augmentation and longer schedules, which can benefit training from scratch, were not evaluated.

Table 3. Native-resolution volume-level test performance under the main protocol (all DINOv2 rows use one FP32 loss implementation). Values are mean ± sample SD across three training seeds. Rows above the first rule use $0.5\mathcal{L}_{\text{BCE}} + 0.5\mathcal{L}_{\text{Dice}}$; bold marks the highest mean within each backbone block, not statistical significance. The controls below use the same split and evaluator: the BCE-only frozen objective, and ViT-L trained from random initialization with the Large Full FT protocol (no DINOv2 weights). nnU-Net uses Dice+CE and a distinct 3D recipe.

Backbone	Adaptation	Dice ↑	IoU ↑
Large	Frozen	0.8747 ± 0.0025	0.7785 ± 0.0040
Large	Partial FT	0.8851 ± 0.0021	0.7953 ± 0.0031
Large	LoRA	0.8992 ± 0.0023	0.8178 ± 0.0038
Large	Full FT	$\mathbf{0.9066} \pm 0.0039$	$\mathbf{0.8301} \pm 0.0065$
Giant	Frozen	0.8753 ± 0.0036	0.7796 ± 0.0056
Giant	Partial FT	0.8911 ± 0.0015	0.8048 ± 0.0023
Giant	LoRA	$\mathbf{0.9027} \pm 0.0005$	$\mathbf{0.8234} \pm 0.0009$
Large	Frozen, BCE only	0.8810 ± 0.0018	0.7887 ± 0.0029
Large	Full FT, random initialization	0.7639 ± 0.0234	0.6426 ± 0.0240
Large	Full FT, random init., LR 10^{-4} + clip	0.7322 ± 0.0084	0.6034 ± 0.0125
nnU-Net ResEncM 3D (reference)		0.9305 ± 0.0001	0.8706 ± 0.0002

5.2. Effect of adaptation strategy

Every adaptation strategy improved mean Dice over the frozen encoder of the same backbone (Table 4, group A). On ViT-Large, Partial FT added 1.05 points, LoRA 2.46, and Full FT 3.19; on ViT-Giant, Partial FT added 1.57 and LoRA 2.73. Under the main protocol, Large Full FT reached the highest DINOv2 Dice and IoU (0.9066 ± 0.0039 and 0.8301 ± 0.0065), although without clipping its validation Dice collapsed after the selected epoch in two of three seeds (Section 5.4); Giant LoRA followed (0.9027 ± 0.0005). Their paired difference was 0.39 points ([0.10, 0.69], $p_{\text{Holm}} = 0.042$; per seed 0.20, 0.88, and 0.10). This gap equals the seed SD of Large Full FT and lies within the hardware retraining variation (0.38 to 0.60 points), and the main Large Full FT runs used a different software stack; it is therefore not interpreted as a strategy effect. Within ViT-Large, Full FT exceeded LoRA by 0.74 points under the main protocol ([0.40, 1.08], unadjusted; per seed 0.86, 1.00, and 0.35), larger than that variation; under the common improved recipe the two tied (Section 5.4), so this gap reflects the recipe rather than an intrinsic advantage of full fine-tuning.

Accuracy did not follow the number of trainable parameters: Large LoRA (1.25 million) exceeded Large Partial FT (51.05 million; 0.8992 versus 0.8851), and on ViT-Giant LoRA exceeded Partial FT by 1.16 points ([0.77, 1.53]).

The nnU-Net reference reached 0.9305 ± 0.0001, 2.39 points above Large Full FT ([1.94, 2.85]) and 2.78 points above Giant LoRA ([2.43, 3.16]); this is a pipeline-level comparison (Section 4.3). In the median- and lowest-ranked test volumes (Figure 2), the randomly initialized encoder departs most from the reference, the adapted models follow the atrial body, and the remaining errors concentrate in narrow extensions of the reference mask, which nnU-Net recovers more completely in the lowest-ranked volume. Coronal reformats (Supplementary Figure S3) show irregular inter-slice boundaries and small detached components in the slice-wise DINOv2 predictions, most visibly for the frozen encoder, whereas the 3D nnU-Net boundary is smoother.

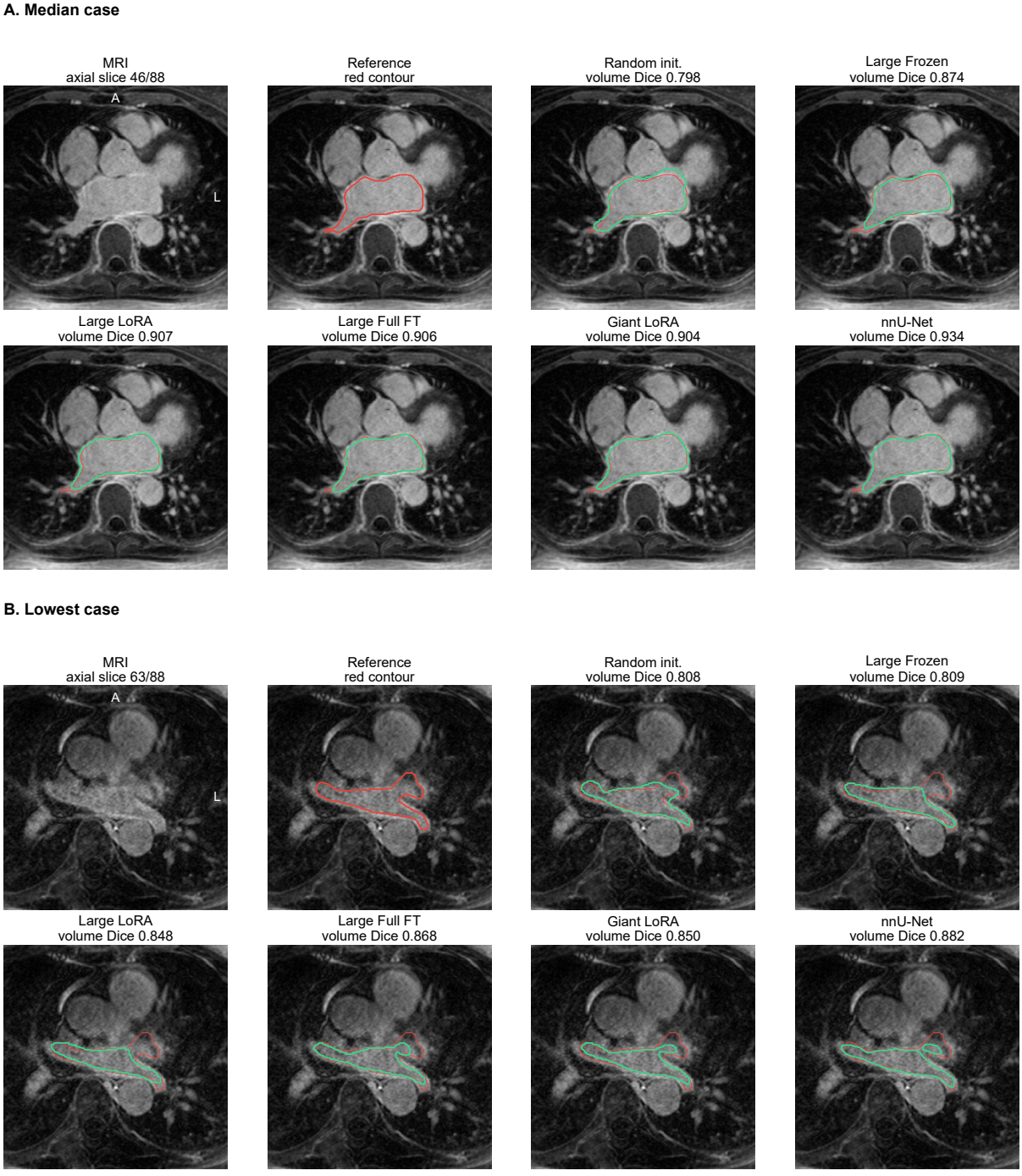


Figure 2. Predictions for (A) the median-ranked and (B) the lowest-ranked test volume, ranked by Dice averaged over the seven main-protocol DINOv2 configurations, the BCE-only frozen control, and three seeds. Each panel shows the axial slice with the largest reference LA area over a field of view that includes the heart, great vessels, and spine (A: anterior; L: patient left). Red: reference; green: prediction (seed 42). Numbers are volume-level Dice of the displayed run, not slice Dice. Random init.: Large Full FT trained from random initialization. Main-protocol models are shown.

Table 4. Paired Dice contrasts (percentage points) for the research questions of the main protocol. For a contrast A − B, positive values indicate higher mean Dice for A; interactions are differences between two Giant − Large scale gains. Seeds are averaged within case. 95% percentile case-bootstrap CIs use 20,000 resamples (RNG seed 20260926, reset for each contrast and subset); p_{Holm} adjusts two-sided bootstrap p values over the 16 contrasts shown. Per-seed differences are given for every contrast because the case bootstrap does not include retraining variance. Intervals are conditional on the observed runs and assume exchangeable test volumes; patient independence is unverified. The 30-case column excludes the identified near-duplicate without retraining.

Comparison	31 cases: Δ [95% CI]	p_{Holm}	Per seed	30 cases: Δ [95% CI]
A. Adaptation versus frozen extraction, same backbone				
Large: Full FT − Frozen	3.19 [2.73, 3.71]	< 0.002	2.77/3.73/3.09	3.20 [2.73, 3.73]
Large: Partial FT − Frozen	1.05 [0.81, 1.32]	< 0.002	0.67/0.99/1.48	1.07 [0.82, 1.35]
Large: LoRA − Frozen	2.46 [2.15, 2.78]	< 0.002	1.91/2.72/2.74	2.52 [2.23, 2.83]
Giant: LoRA − Frozen	2.73 [2.40, 3.14]	< 0.002	2.54/2.51/3.14	2.74 [2.39, 3.16]
Giant: Partial FT − Frozen	1.57 [1.22, 1.96]	< 0.002	1.39/1.23/2.10	1.46 [1.15, 1.76]
B. Backbone scale, Giant − Large, same strategy				
Frozen	0.07 [−0.22, 0.35]	0.626	0.02/0.34/−0.16	0.12 [−0.15, 0.40]
LoRA	0.35 [0.16, 0.54]	< 0.002	0.66/0.13/0.25	0.35 [0.16, 0.54]
Partial FT (last four blocks)	0.59 [0.30, 0.93]	< 0.002	0.75/0.57/0.46	0.51 [0.24, 0.81]
C. Scale interaction: difference between scale gains				
LoRA − Frozen	0.28 [0.03, 0.55]	0.045	0.64/−0.21/0.41	0.22 [−0.003, 0.48]
Partial FT − Frozen	0.53 [0.11, 0.98]	0.042	0.73/0.24/0.62	0.39 [0.02, 0.75]
D. Practical configuration comparisons				
Large Full FT − Giant LoRA	0.39 [0.10, 0.69]	0.042	0.20/0.88/0.10	0.34 [0.05, 0.62]
Giant LoRA − Giant Partial FT	1.16 [0.77, 1.53]	< 0.002	1.15/1.29/1.04	1.28 [1.00, 1.59]
Large LoRA − Giant Partial FT	0.81 [0.39, 1.21]	< 0.002	0.49/1.16/0.80	0.94 [0.58, 1.27]
E. Task-specific reference with a distinct recipe				
nnU-Net − Large Full FT	2.39 [1.94, 2.85]	< 0.002	2.54/1.93/2.69	2.44 [2.00, 2.90]
nnU-Net − Giant LoRA	2.78 [2.43, 3.16]	< 0.002	2.74/2.81/2.79	2.78 [2.42, 3.18]
nnU-Net − Giant Partial FT	3.94 [3.40, 4.56]	< 0.002	3.89/4.10/3.83	4.07 [3.57, 4.65]

All DINOv2 rows use the BCE–Dice objective and one loss implementation. Partial FT updates the last four blocks, not a matched fraction of depth. nnU-Net uses its distinct 3D recipe. Per-seed values follow seed order 42/2026/3407.

5.3. Effect of backbone scale

Replacing Large with Giant changed Dice by 0.07 points for the frozen encoder ([−0.22, 0.35]), 0.35 points for LoRA ([0.16, 0.54]), and 0.59 points for Partial FT ([0.30, 0.93]; Table 4, group B; Figure 3). A larger encoder thus gave no measurable gain when frozen and a small gain when adapted, and every adapted Large configuration exceeded Giant Frozen. The two scale interactions (LoRA 0.28 points, [0.03, 0.55]; Partial FT 0.53 points, [0.11, 0.98]) had Holm-adjusted p values of 0.045 and 0.042; the LoRA interaction fails the robustness criterion (its sign reversed for one seed), whereas the Partial FT interaction meets it; because both are secondary contrasts with wide intervals, we treat both as exploratory. Because Giant Full FT was not evaluated, the comparison of Large Full FT with Giant LoRA changes both strategy and scale and cannot isolate either factor. Supplementary Figures S1 and S2 show all contrasts and their per-case distributions.

5.4. Recipe sensitivity

Table 5 reports each single factor within the strategy to which it was applied. The four-layer readout had the largest effect, raising Dice by 2.24 points for Large Frozen (30 of 31 cases higher) and 0.73 points for Large LoRA (28 of 31) without extra trainable parameters. The higher adapter learning rate raised Large LoRA by 0.46 points ([0.20, 0.75]; higher for all seeds) and Giant LoRA by 0.26 points ([0.001, 0.51]; a cross-hardware contrast within retraining variation).

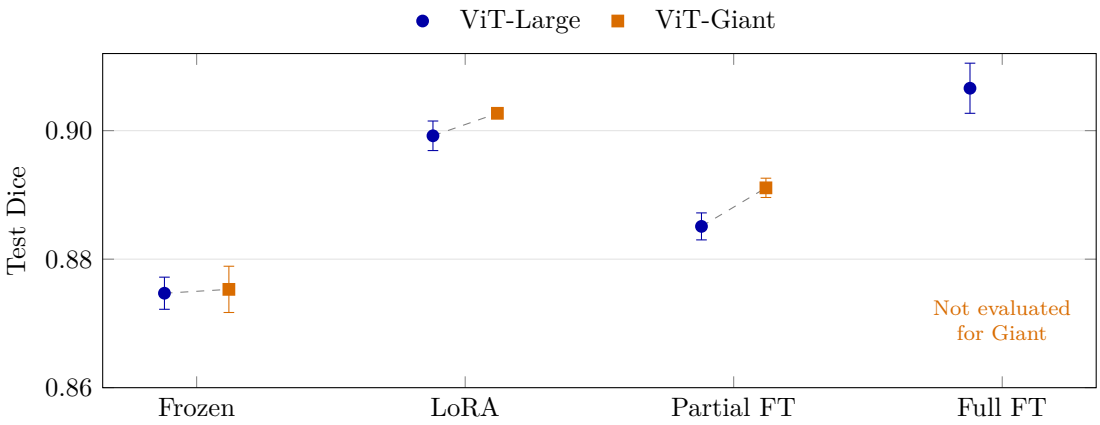


Figure 3. Observed strategy–scale matrix under the main protocol. Markers are three-seed mean Dice and error bars are sample SD across seeds, not confidence intervals. Dashed lines connect matched Large–Giant comparisons. Giant Partial FT uses the last four blocks, matching block count rather than depth fraction. Giant Full FT was not evaluated; paired CIs are given in Table 4.

Table 5. Recipe sensitivity within each strategy (RQ3; post hoc, three seeds, selection on validation only). Each factor was varied for the strategy shown; rows are not compared across strategies. Δ is the paired case-bootstrap difference (percentage points, unadjusted 95% CI) against the reference stated in the row; per-seed differences follow seed order 42/2026/3407.

Factor (strategy)	Reference	Variant	Δ [95% CI]	Cases > 0	Per seed
Adapter LR 10^{-4} (Large LoRA)	0.8992 ± 0.0023	0.9038 ± 0.0009	0.46 [0.20, 0.75]	24/31	0.63/0.38/0.36
Adapter LR 10^{-4} (Giant LoRA)	0.9027 ± 0.0005	0.9053 ± 0.0007	0.26 [0.001, 0.51]	23/31	0.28/0.31/0.20
Giant − Large at LR 10^{-4} (LoRA)	0.9038 ± 0.0009	0.9053 ± 0.0007	0.15 [−0.004, 0.30]	21/31	0.30/0.06/0.09
Gradient clipping 1.0 (Large Full FT)	0.9066 ± 0.0039	0.9125 ± 0.0006	0.59 [0.41, 0.77]	27/31	0.67/0.18/0.92
Four-layer readout (Large Frozen)	0.8757 ± 0.0007	0.8981 ± 0.0012	2.24 [1.80, 2.67]	30/31	2.20/2.22/2.29
Four-layer readout (Large LoRA)	0.8971 ± 0.0027	0.9044 ± 0.0022	0.73 [0.47, 0.98]	28/31	0.51/0.87/0.81

The adapter learning rate was selected on validation in a single-seed sweep of Large LoRA (10^{-5}, 10^{-4}; 10^{-3} diverged) and applied unchanged to Giant LoRA. Clipped Full FT runs used an RTX 5090 and the reference runs an RTX PRO 6000; Giant LoRA reference runs used an RTX 4000 Ada and its 10^{-4} runs an RTX 4090. An unclipped seed-42 rerun on the RTX 5090 reached Dice 0.9064, against 0.9119 with clipping (its validation MDD is in Table S7). Readout cells are independent retrainings with the main protocol otherwise unchanged; seeds 42 and 2026 used an RTX 4090 and seed 3407 an RTX 5090, with the four cells of a seed on one GPU. The four-layer readout partitions the original projection and adds no trainable parameters.

This reduced the LoRA scale gain from 0.35 to 0.15 points ([−0.004, 0.30]) and the advantage of unclipped Large Full FT over Giant LoRA from 0.39 to 0.13 points ([−0.07, 0.35]). Gradient clipping prevented the validation collapse of unclipped Large Full FT, whose validation Dice had fallen to near zero in two of three seeds after strong checkpoints (MDD 0.89, 0.51, and 0.89; Table S7); with clipping it stayed at or above 0.865 after epoch 3, and the seed SD of test Dice fell from 0.0039 to 0.0006. An unclipped rerun on the GPU model of the clipped runs again declined transiently (MDD 0.63). Under these single-factor changes, Full FT stayed ahead of LoRA: clipped Large Full FT exceeded Large and Giant LoRA at the higher adapter rate by 0.88 points ([0.64, 1.15]) and 0.72 points ([0.48, 1.00]).

Table 6 applies the three factors together to all four ViT-Large strategies on one hardware stack. Every strategy improved, and the gains were largest for the weakest strategies: 2.86 points for the frozen encoder, 2.28 for Partial FT, 1.23 for LoRA, and 0.49 for Full FT, with the same sign in all seeds except one Full FT seed (−0.02). LoRA and Full FT reached the same mean Dice (0.9115 ± 0.0007 and 0.9115 ± 0.0011; difference 0.000 points, [−0.19, 0.18]), and the gap

Table 6. Common improved protocol for the four ViT-Large strategies (post hoc; three seeds; all runs on one RTX 5090 and software stack). Every strategy uses the four-layer readout, gradient-norm clipping at 1.0 and the learning-rate rule (pretrained weights 10^{-5}, LoRA adapters 10^{-4}, decoder 10^{-3}); all other settings follow the main protocol, and the test set was evaluated only after all 12 runs had finished. Δ: paired case-bootstrap difference in percentage points (unadjusted 95% CI); per-seed differences follow seed order 42/2026/3407.

Strategy (ViT-Large)	Main protocol	Common improved	Δ [95% CI]	Cases > 0	Per seed
Frozen	0.8747 ± 0.0025	0.9032 ± 0.0008	2.86 [2.44, 3.32]	31/31	2.63/2.98/2.96
Partial FT	0.8851 ± 0.0021	0.9079 ± 0.0010	2.28 [1.90, 2.73]	31/31	2.48/2.33/2.04
LoRA	0.8992 ± 0.0023	0.9115 ± 0.0007	1.23 [0.95, 1.53]	30/31	1.41/1.10/1.18
Full FT	0.9066 ± 0.0039	0.9115 ± 0.0011	0.49 [0.31, 0.68]	27/31	0.76/−0.02/0.73
Contrasts under the common improved protocol					
Full FT − LoRA			0.000 [−0.19, 0.18]	18/31	0.21/−0.11/−0.10
LoRA − Partial FT			0.36 [0.12, 0.63]	23/31	0.17/0.50/0.40
Partial FT − Frozen			0.47 [0.21, 0.75]	22/31	0.52/0.34/0.55
Full FT − Frozen			0.83 [0.55, 1.11]	28/31	0.90/0.73/0.86

between Full FT and the frozen encoder fell from 3.19 to 0.83 points ([0.55, 1.11]). LoRA still exceeded Partial FT (0.36 points, [0.12, 0.63]) and Partial FT the frozen encoder (0.47 points, [0.21, 0.75]), with the same sign in all seeds. Peak allocated memory recorded in these runs (1.7, 2.9, 4.9, and 8.8 GB for Frozen, Partial FT, LoRA, and Full FT) kept the main-protocol ordering, so LoRA reached the Dice of Full FT with about 56% of its memory, but not in less estimated training time, because it trained for more epochs (about 2.0 versus 1.7 GPU-hours; Table S7).

5.5. Reduced label budgets

Table 7 repeats the common improved recipe with fewer training volumes. The contribution of pretraining grew as labels were reduced: Full FT from DINOv2 initialization exceeded the randomly initialized encoder by 14.99 points with 108 volumes, 26.85 with 27, and 41.95 with 11 ([36.55, 47.66]; higher in all 31 cases). The randomly initialized encoder lost 30.01 points between 108 and 11 volumes and the pretrained strategies 2.84 to 4.22 points, so pretrained Full FT trained on 11 volumes still exceeded the randomly initialized encoder trained on all 108, by 11.94 points ([7.86, 17.26]; across budgets, unadjusted). The comparator is an unaugmented ViT trained from scratch with a recipe not tuned for small training sets, so these magnitudes are likely upper-range estimates of the pretraining effect. LoRA and Full FT did not differ clearly at any budget (0.22 points with 11 volumes, [−0.17, 0.58]; 0.37 with 27, [−0.24, 1.22]), although differences of about one point cannot be excluded with 27 volumes, whereas the advantage of LoRA over the frozen encoder grew from 0.83 to 2.21 points ([1.40, 3.19]). With 11 volumes, Full FT selected its checkpoint after one or two epochs (each about ten passes over the subset) in all seeds, and LoRA after 9 to 23.

nnU-Net lost 1.02 points ([0.65, 1.44]) between 108 and 11 volumes, less than any DINOv2 configuration, so its advantage over LoRA widened from 1.90 to 3.71 points ([3.05, 4.42]; change 1.82, [1.30, 2.39]), with higher nnU-Net Dice in all 31 cases at 11 volumes. In an exploratory comparison across budgets, nnU-Net trained on 11 volumes also exceeded LoRA and Full FT trained on all 108, by 0.88 points ([0.27, 1.43] and [0.26, 1.44]), a margin of the order of the retraining variation across GPUs. Every contrast whose interval excluded zero had the same sign in all three seeds, and the case × seed bootstrap, which also reflects subset variation, excluded zero for the same contrasts (for example, [36.13, 49.33] for the 11-volume pretraining gain and [1.00, 2.69] for the widening of the nnU-Net advantage). Under this protocol, fewer

Table 7. Reduced label budgets (RQ1; exploratory, design fixed before any reduced-data run but after the main results). ViT-Large configurations use the common improved recipe without augmentation and nnU-Net its own recipe with augmentation; each seed trains on one nested random subset of the 108 training volumes, shared by all configurations, and every run selects its checkpoint on the same 15 annotated validation volumes and is tested on the same 31 volumes. Random init.: the same ViT-Large trained from random initialization with the common recipe and an encoder learning rate of 10^{-4}; its 108-volume runs are new and therefore differ from both from-scratch rows of Table 3. The other 108-volume DINOv2 rows are the runs of Table 6. Upper block: Dice, mean ± SD of three seed means. Lower block: paired case-bootstrap differences in percentage points with 95% CIs; the last column is the change of each quantity from 108 to 11 training volumes.

Training volumes	11 (10%)	27 (25%)	108 (100%)	Change, 11 − 108
Dice				
Random init. (Full FT)	0.4615 ± 0.0386	0.6260 ± 0.0374	0.7616 ± 0.0070	−30.01 [−34.40, −25.91]
Frozen	0.8611 ± 0.0088	0.8837 ± 0.0018	0.9032 ± 0.0008	−4.22 [−5.49, −3.11]
Partial FT	0.8724 ± 0.0064	0.8859 ± 0.0007	0.9079 ± 0.0010	−3.56 [−4.38, −2.81]
LoRA	0.8832 ± 0.0059	0.8981 ± 0.0012	0.9115 ± 0.0007	−2.84 [−3.54, −2.18]
Full FT	0.8810 ± 0.0056	0.8945 ± 0.0013	0.9115 ± 0.0011	−3.05 [−3.65, −2.50]
nnU-Net ResEncM 3D	0.9203 ± 0.0008	0.9246 ± 0.0017	0.9305 ± 0.0001	−1.02 [−1.44, −0.65]
Paired differences, percentage points				
Full FT − random init.	41.95 [36.55, 47.66]	26.85 [22.17, 32.30]	14.99 [10.74, 20.46]	26.96 [22.99, 31.17]
LoRA − Full FT	0.22 [−0.17, 0.58]	0.37 [−0.24, 1.22]	0.000 [−0.18, 0.19]	0.22 [−0.19, 0.58]
LoRA − Frozen	2.21 [1.40, 3.19]	1.45 [0.97, 2.05]	0.83 [0.55, 1.11]	1.38 [0.64, 2.25]
nnU-Net − LoRA	3.71 [3.05, 4.42]	2.65 [2.23, 3.10]	1.90 [1.54, 2.26]	1.82 [1.30, 2.39]
nnU-Net − Full FT	3.93 [3.31, 4.62]	3.01 [2.24, 4.07]	1.90 [1.49, 2.32]	2.03 [1.56, 2.54]

p_{Holm} over the 15 contrasts at 11 and 27 volumes and in the change column (family defined after these results were viewed): LoRA − Full FT, 11 volumes 0.782, 27 volumes 0.782, change 0.782; all others < 0.002. The 108-volume contrasts carry unadjusted intervals. Every contrast whose interval excludes zero has the same sign in all three seeds, and a case × seed bootstrap that also resamples the subsets excludes zero for the same contrasts and only for them. Every reduced-data epoch draws 2,376 batches, as a full-data epoch does, so all budgets share the update schedule and early-stopping rule. nnU-Net plans were derived from the 123 development volumes for every budget.

labels thus enlarged both the benefit of pretraining over random initialization and the gap to the task-specific pipeline, which alone used augmentation and 3D context.

5.6. Decoder capacity

Table 8 replaces the lightweight decoder of the common improved recipe with the hierarchical decoder. With 108 training volumes, the hierarchical decoder raised Dice by 0.50 points for the frozen encoder ([0.25, 0.75]) and by 0.33 points for Full FT ([0.20, 0.46]), with the same sign in all seeds, whereas the changes for Partial FT (0.13 points) and LoRA (0.09 points) did not exclude zero. Full FT with the hierarchical decoder reached the highest DINOv2 Dice of the study (0.9148 ± 0.0017) and exceeded LoRA with the same decoder by 0.24 points ([0.02, 0.47]; same sign in all seeds, but [−0.04, 0.52] in the case × seed bootstrap), so the ordering by mean Dice changed only between these two strategies, which had tied with the lightweight decoder; the frozen encoder came within 0.66 points of Full FT. With 11 training volumes, the hierarchical decoder lowered Dice instead: by 2.75 points for Full FT ([−4.59, −1.38]; all seeds), whose validation Dice was also lower (0.837 to 0.844, versus 0.852 to 0.862 with the lightweight decoder), and by 0.68 points for LoRA, a decrease that came from one seed (per seed −2.30, 0.15, and 0.12) and therefore fails the robustness criterion. nnU-Net remained above every hierarchical-decoder configuration, by 1.57 points over Full FT with 108 volumes ([1.23, 1.92]; 1.90 with the lightweight decoder) and by 4.39 and 6.68 points over LoRA and Full FT with 11 volumes. Decoder capacity thus changed Dice less than the readout and recipe factors of Section 5.4, narrowed the gap between the best DINOv2 configuration and nnU-Net from 1.90 to 1.57 points, and did not help with 11 training volumes.

Table 8. Decoder capacity (RQ3; post hoc with respect to the main and common-recipe results, with design and planned contrasts fixed before any of these runs). ViT-Large strategies with the common improved recipe and either the lightweight decoder with the four-layer readout (0.66 million parameters; the runs of Tables 6 and 7) or a hierarchical decoder with skip connections from encoder blocks 5, 11, 17, and 23 and from the input slice (3.12 million parameters). Subsets, batch order, learning rates, checkpoint rule, and evaluation are identical, and all hierarchical-decoder runs used one RTX 5090. Dice: mean ± SD of three seed means. Δ: hierarchical minus lightweight decoder. Differences are paired case-bootstrap differences in percentage points with 95% CIs; per-seed differences follow seed order 42/2026/3407.

Strategy	Lightweight	Hierarchical	Δ [95% CI]	Per seed	nnU-Net 3D − hierarchical
108 training volumes (nnU-Net 3D: 0.9305 ± 0.0001)					
Frozen	0.9032 ± 0.0008	0.9082 ± 0.0015	0.50 [0.25, 0.75]	0.34/0.39/0.76	2.23 [1.79, 2.73]
Partial FT	0.9079 ± 0.0010	0.9092 ± 0.0033	0.13 [−0.11, 0.37]	0.25/0.39/−0.25	2.12 [1.79, 2.48]
LoRA	0.9115 ± 0.0007	0.9124 ± 0.0021	0.09 [−0.15, 0.31]	0.30/−0.19/0.16	1.81 [1.51, 2.11]
Full FT	0.9115 ± 0.0011	0.9148 ± 0.0017	0.33 [0.20, 0.46]	0.35/0.20/0.44	1.57 [1.23, 1.92]
11 training volumes (nnU-Net 3D: 0.9203 ± 0.0008)					
LoRA	0.8832 ± 0.0059	0.8764 ± 0.0171	−0.68 [−1.25, −0.19]	−2.30/0.15/0.12	4.39 [3.52, 5.43]
Full FT	0.8810 ± 0.0056	0.8535 ± 0.0083	−2.75 [−4.59, −1.38]	−2.90/−2.97/−2.37	6.68 [4.94, 9.00]
Contrasts with the hierarchical decoder, 108 training volumes (unadjusted)					
Full FT − LoRA			0.24 [0.02, 0.47]	0.26/0.29/0.18	
LoRA − Partial FT			0.32 [0.13, 0.52]	0.22/−0.08/0.82	
Partial FT − Frozen			0.10 [−0.12, 0.33]	0.43/0.33/−0.45	
Full FT − Frozen			0.66 [0.37, 0.96]	0.91/0.54/0.54	

p_{Holm} over the 12 planned contrasts: Δ for LoRA with 11 volumes 0.016, Partial FT with 108 volumes 0.560, LoRA with 108 volumes 0.560; all others < 0.002. Every planned contrast whose interval excludes zero has the same sign in all three seeds, except Δ for LoRA with 11 volumes; a case × seed bootstrap excludes zero for the other contrasts with intervals excluding zero and for no further contrast.

5.7. External evaluation

Table 9 applies the main-protocol and common-recipe models to external data without retraining. On the 40 CDEMRIS LGE volumes from 20 patients, every DINOv2 configuration had higher mean Dice than nnU-Net (0.759–0.784 versus 0.755), but no paired difference excluded zero and the median per-case differences ranged from −1.7 to +0.4 points, so the comparison is inconclusive; the means were dominated by one volume in which nnU-Net failed for all seeds (Dice 0.02, versus 0.78 to 0.87 for the DINOv2 configurations), although no preprocessing difference from the other volumes was found. Without that volume (post hoc), the differences ranged from −1.5 to +0.9 points, again with every interval including zero. The internal nnU-Net advantage over Large Full FT (2.39 points) was therefore not observed externally (Large Full FT minus nnU-Net: 2.1 points, [−0.9, 6.7]; 0.1 points without that volume), although this sample does not establish equivalence. On the 20 MSD bSSFP volumes, every DINOv2 configuration was lower than nnU-Net, from −7.4 points ([−11.3, −3.0]) for Giant LoRA to −19.9 points for Large Frozen, and the BCE-only frozen encoder (0.607) exceeded several adapted configurations, so the external ordering of strategies also depends on the objective. The common recipe did not change this pattern: its four models were 2.0 to 3.0 points above nnU-Net on CDEMRIS and 8.9 to 13.4 points below it on MSD. Its LoRA and Full FT models, tied internally, differed in opposite directions on the two datasets, with neither interval excluding zero (Full FT minus LoRA: −0.9 points on CDEMRIS, [−2.4, 0.1]; 4.5 points on MSD, [−0.02, 9.4]).

5.8. Accuracy–resource trade-offs

Parameter efficiency did not translate into proportional memory or time savings (Table 10, Figure 4). Within ViT-Large, LoRA needed 4.9 GB of peak training memory versus 8.7 GB for Full FT and 9.6 versus 14.1 seconds per 100 fixed-workload batches, at 0.74 points lower Dice

Table 9. External evaluation without retraining (RQ4, exploratory). CDEMRIS: 40 LGE volumes, one pre- and one post-ablation scan from each of 20 patients, two non-Utah centers; MSD Task02: the 20 bSSFP volumes with public reference masks (different sequence). Dice is the mean ± SD of three seed means; Δ is the paired difference configuration minus nnU-Net in percentage points (positive values favor the configuration) with an unadjusted 95% bootstrap CI that resamples patients (CDEMRIS) or volumes (MSD). Label definitions across centers may differ. The common-recipe rows were scored post hoc with the same external protocol.

Configuration	CDEMRIS Dice	Δ vs nnU-Net	MSD Dice	Δ vs nnU-Net
Large Frozen (BCE)	0.765 ± 0.004	1.0 [−2.0, 5.5]	0.607 ± 0.016	−11.9 [−14.4, −9.3]
Large Frozen	0.759 ± 0.005	0.4 [−2.8, 4.9]	0.528 ± 0.032	−19.9 [−22.9, −16.6]
Giant Frozen	0.762 ± 0.001	0.7 [−2.3, 5.2]	0.538 ± 0.018	−18.9 [−21.9, −15.6]
Large Partial FT	0.772 ± 0.004	1.7 [−1.3, 6.3]	0.553 ± 0.022	−17.4 [−23.1, −11.8]
Giant Partial FT	0.775 ± 0.004	2.0 [−0.9, 6.6]	0.560 ± 0.052	−16.6 [−20.8, −12.8]
Large LoRA	0.770 ± 0.001	1.5 [−1.4, 6.0]	0.587 ± 0.033	−14.0 [−19.1, −8.8]
Giant LoRA	0.779 ± 0.003	2.4 [−0.6, 6.9]	0.653 ± 0.027	−7.4 [−11.3, −3.0]
Large Full FT	0.776 ± 0.009	2.1 [−0.9, 6.7]	0.640 ± 0.022	−8.7 [−13.2, −3.5]
Common improved recipe (ViT-Large; post hoc)				
Frozen	0.775 ± 0.001	2.0 [−0.9, 6.5]	0.630 ± 0.028	−9.6 [−13.3, −6.0]
Partial FT	0.779 ± 0.001	2.4 [−0.7, 7.2]	0.625 ± 0.026	−10.1 [−14.0, −6.3]
LoRA	0.784 ± 0.000	3.0 [−0.1, 7.5]	0.592 ± 0.049	−13.4 [−19.2, −8.3]
Full FT	0.775 ± 0.006	2.0 [−1.0, 6.7]	0.637 ± 0.029	−8.9 [−14.3, −4.0]
nnU-Net ResEncM 3D	0.755 ± 0.015	–	0.726 ± 0.019	–

Common recipe, Full FT − LoRA (percentage points, unadjusted 95% CI): CDEMRIS −0.9 [−2.4, 0.1], MSD 4.5 [−0.02, 9.4].

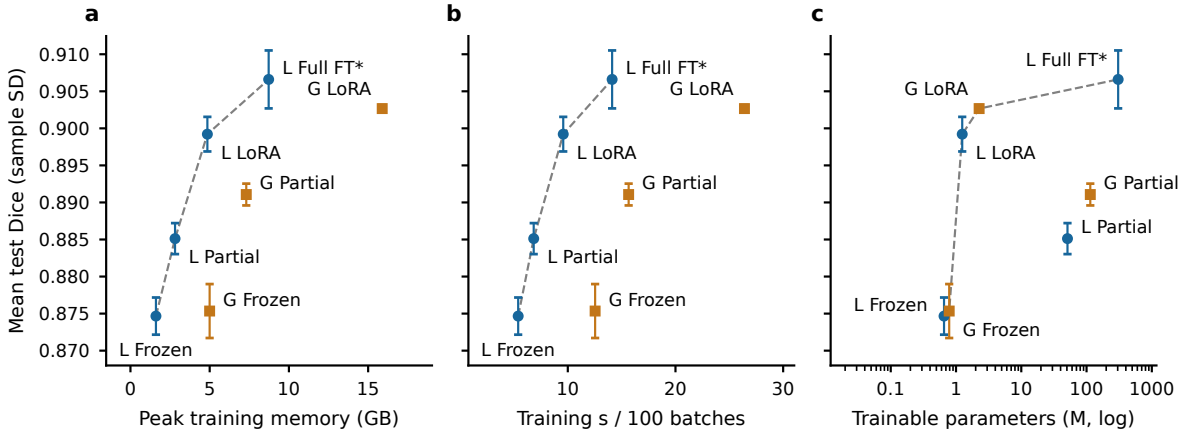


Figure 4. Observed trade-offs between accuracy and (a) peak training memory, (b) training time, and (c) trainable parameters under the main protocol. Dice is the three-seed mean with sample SD bars; resources come from the fixed workload (batch 4, 448 by 448, RTX 4090, three repeats of 100 batches). Dashed lines mark the observed nondominated points in each panel. L and G denote ViT-Large and ViT-Giant; FT, fine-tuning. Large Full FT (*) had one skipped optimizer update during the benchmark.

under the main protocol and equal Dice under the common recipe; their estimated total GPU-hours were similar (2.04 and 2.08) because LoRA trained for more epochs. Across backbones, Giant LoRA updated about 135 times fewer parameters than Large Full FT but required 15.9 versus 8.7 GB, 26.4 versus 14.1 seconds per 100 batches, and the largest estimated training cost (5.5 GPU-hours). Because Giant configurations added at most 0.59 points over their Large counterparts, the larger backbone mainly increased cost.

Table 10. Trainable parameters, peak allocated training memory and estimated training cost under the main protocol. Peak memory is the largest allocated peak of the fixed-workload benchmark (Table S5). GPU-hours are estimates on the RTX 4090 reference: measured fixed-workload seconds per training batch × 2,376 batches per epoch × epochs completed, in total or up to the validation-selected checkpoint; mean ± SD across seeds, where the SD reflects only the number of epochs completed before early stopping. Validation, checkpoint writing and test inference are excluded.

Backbone	Adaptation	Trainable parameters	Peak memory (GB)	GPU-hours (total)	GPU-hours (to best)
Large	Frozen	655,329	1.609	1.08 ± 0.14	0.72 ± 0.14
Large	Partial FT	51,048,417	2.814	1.10 ± 0.36	0.65 ± 0.36
Large	LoRA	1,245,153	4.854	2.04 ± 0.97	1.54 ± 1.19
Large	Full FT	305,023,969	8.722	2.08 ± 0.48	1.15 ± 0.48
Giant	Frozen	786,401	4.997	1.96 ± 0.53	1.13 ± 0.53
Giant	Partial FT	114,132,961	7.309	4.61 ± 0.39	3.58 ± 0.39
Giant	LoRA	2,260,961	15.888	5.52 ± 0.79	3.78 ± 0.79

The BCE-only frozen control has the Large Frozen architecture (655,329 trainable parameters, 1.609 GB); using the Large Frozen throughput, its estimated cost is 0.89 ± 0.13 GPU-hours in total and 0.53 ± 0.13 to the selected checkpoint.

5.9. Seed variation and sensitivity to the near-duplicate case

Among the main-protocol DINOv2 configurations, Giant LoRA had the smallest seed SD of Dice (0.0005) and unclipped Large Full FT the largest (0.0039; Table S2). Excluding the test volume identified as a near-duplicate of a training volume lowered mean Dice by 0.12 to 0.24 points for every main-protocol configuration and left their ordering unchanged (Table S3). Apart from that pair (coarse-grid image correlation 0.999; 0.996 at native resolution), test volumes correlated at most 0.80 with any training volume on the coarse grid (median 0.64). On the 21 least similar test volumes, every key contrast kept its sign and a similar size, including the 2.39-point nnU-Net advantage and the tie between Full FT and LoRA under the common recipe, and per-case differences were only weakly related to similarity ($|r| \leq 0.37$; Table S4). The case × seed bootstrap kept every key interval away from zero except the Large Full FT versus Giant LoRA gap ($[-0.07, 0.88]$) and, as expected for equal means, the common-recipe Full FT versus LoRA contrast ($[-0.26, 0.25]$); the former gap and the scale gains are of the order of hardware retraining variation.

6. Discussion

6.1. What the controlled comparison shows

Under one protocol and within the evaluated ranges, pretraining mattered most (14.3 points against a ViT trained from scratch without augmentation, and 42.0 points against the same unaugmented comparator with 11 training volumes), followed by encoder adaptation and decoder readout (about 1 to 3 points), backbone scale (at most 0.6 points), and decoder capacity (at most 0.5 points). These magnitudes depend on the levels compared, so they describe this protocol rather than a general ranking of factors. The common improved recipe shows how strongly the strategy gaps depend on training settings: once every ViT-Large strategy received the same readout, clipping, and learning-rate rule, LoRA reached the mean Dice of full fine-tuning and the frozen encoder came within 0.83 points. Most of the gaps under the main protocol therefore reflected recipe settings rather than intrinsic limits of constrained adaptation. With 11 training volumes, LoRA and full fine-tuning still did not differ clearly, but the frozen encoder fell 2.2 points behind, so adapting the encoder mattered more when labels were scarce. Method rankings are known to depend on evaluation choices [17]; here they also depended on training choices, and in a SAM benchmark the advantage of parameter-efficient over full fine-tuning likewise depended on backbone size [13].

LoRA outperformed partial fine-tuning internally under both protocols despite updating about 41 times fewer parameters on ViT-Large. A plausible but untested explanation is that low-rank updates distributed across all blocks permit adjustments that are unavailable when only the final blocks change. The readout results suggest that much of the frozen-versus-adapted gap under the main protocol came from the last-layer decoder rather than from the encoder update. Multi-level features and stronger decoders have also improved DINOv2 segmentation in other studies [6, 7, 10, 12], and a broader adaptation benchmark emphasized the role of the decoder [15]. Here, once the readout drew on several encoder blocks, a hierarchical decoder with skip connections added at most 0.5 points with all training volumes and lowered Dice with 11 (Section 5.6).

Because the scale gains were small and recipe-dependent, a larger encoder offered little return at its higher cost, as larger SAM backbones did in another benchmark [13]. Without clipping, the best validation checkpoint of Full FT can precede a collapse, so the unclipped Full FT result is conservative for this strategy; degraded full fine-tuning with a newly initialized decoder has also been attributed to unstable gradients [15].

6.2. Relation to prior work and to task-specific pipelines

The findings complement prior DINOv2-based LA work [9, 10, 12] by separating adaptation strategy, backbone scale, and pretraining under a common decoder topology. Kundu et al. reported DINOv2 with U-Net-style decoders above nnU-Net, including on the 2018 challenge data [10], whereas here the 3D nnU-Net reference remained at least 1.57 points above every DINOv2 configuration, and further above with fewer training volumes. The designs differ: their decoders use skip connections, multi-scale fusion, or image input, our main decoder is a lightweight decoder shared across encoders, their nnU-Net reference was 2D in one study [12], and partitions and preprocessing differ. Neither the decoder nor the 2D formulation closed the gap here. Multi-level features raised frozen DINOv2 by 2.2 points, and a hierarchical decoder with skip connections from four encoder depths and the input slice narrowed the gap to nnU-Net only from 1.90 to 1.57 points (Section 5.6). A 2D nnU-Net trained with the reference recipe reached 0.9233 ± 0.0005, 0.72 points below the 3D reference ([0.40, 1.06]) but still 0.84 points above the best DINOv2 configuration ([0.52, 1.17]; Supplementary Table S9), so slice-wise processing explains part, but not all, of the gap.

The no-pretraining control, the label-budget runs, and the nnU-Net reference together contextualize the role of pretraining. For this ViT pipeline, natural-image pretraining is what makes the transformer encoder competitive, and more so the fewer volumes are annotated, yet a convolutional network trained from scratch with 3D context, augmentation, deep supervision, and a longer schedule reached the highest internal Dice at every budget and lost the least when labels were reduced. This pattern is consistent with pretraining offsetting the data demands of a large ViT rather than making this pipeline more label-efficient than the task-specific one, an interpretation this study did not test directly: relative to random initialization, pretraining reduced the annotation needed, as anticipated [7, 9], since pretrained Full FT on 11 volumes outperformed the randomly initialized encoder on 108, but not relative to nnU-Net, which with 11 volumes exceeded every DINOv2 configuration trained on 108. This comparison is pipeline-level, because the DINOv2 models were trained without augmentation, which in nnU-Net helped most for very small training sets [36]. A similar pattern was reported for brain MRI, where fine-tuned SAM and MedSAM did not outperform a standard U-Net even with very few training volumes [37]. The result also agrees with the small benefit of ImageNet transfer reported for lightweight convolutional models on large medical datasets [16] and with the observation that strongly configured task-specific baselines remain difficult to surpass [5].

Externally, the comparison with nnU-Net was inconclusive on LGE data from other centers, whereas the DINOv2 models were clearly lower on bSSFP data; the reason was not isolated, because dimensionality, normalization, and augmentation differ between the pipelines. The common improved recipe changed neither finding, and neither result supports a general robustness advantage of the pretrained encoder.

6.3. Practical recommendations

Table 11 condenses the evidence into recommendations for practitioners who reuse a DINOv2 encoder for LA segmentation, each tied to the result that supports it and to its boundary. They apply to the evaluated protocol and data size and do not replace validation on a new dataset.

6.4. Limitations

The most important limitation is possible patient overlap between training and test data. The collection contains pre- and post-ablation scans, but the split is case-level without a verified patient mapping and could not be redone by patient; the similarity screen detects near-duplicates rather than patient identity, and undetected repeat scans of a patient cannot be excluded. Internal Dice values may therefore be optimistic and should not be read as patient-level generalization, although the comparisons between configurations were stable on the least similar test volumes. The external evaluation uses patients from other centers but is inference-only, small (40 LGE volumes from 20 patients and 20 bSSFP volumes), and may involve different label definitions.

The strategy comparisons hold for a shared protocol with nominally fixed hyperparameters. All recipe analyses were post hoc; the common improved recipe is one recipe rather than a per-strategy optimization and covers ViT-Large only, and the label-budget and decoder models were not evaluated externally. Training on heterogeneous shared GPUs adds retraining variation of a few tenths of a point, and the main Large Full FT runs used a different software stack from the other main-table rows. The decoder comparison covers one hierarchical design on ViT-Large with the schedule of the common recipe; its decrease with 11 volumes may reflect this untuned schedule rather than a limit of decoders with skip connections. The label-budget analysis drew one nested subset per seed, so subset and retraining variation are only jointly resampled; it covers two reduced budgets and ViT-Large only; no schedule, regularization, or augmentation was tuned per budget, which may affect the from-scratch and Full FT runs most; all budgets kept the 15 annotated validation volumes, more than the smallest training subset, so

Table 11. Practical recommendations derived from this study. Effects are paired differences in Dice percentage points on the internal test set unless stated otherwise. Rows 2–4 and 7 and all label-budget and decoder evidence rest on post hoc analyses (rows 2 and 7 and the label budgets on the common improved recipe for ViT-Large) and are provisional.

Recommendation	Supporting evidence and boundary
Initialize the ViT encoder from DINOv2 rather than training it from scratch.	+14.3 points over random initialization, all 31 cases; a second from-scratch recipe (higher learning rate, clipping) was lower. The gain grew to +42.0 points with 11 training volumes, against the same unaugmented comparator. Augmentation and long schedules for scratch training were not tested.
Prefer Large LoRA with a four-layer readout, gradient clipping, and an adapter learning rate set separately from the pretrained weights.	Same mean Dice as Large Full FT under the common improved recipe (0.9115 each; difference 0.000, [−0.19, 0.18]) with about 56% of its training memory (4.9 versus 8.8 GB), but not less estimated training time; no clear difference with 11 or 27 training volumes either, although about one point cannot be excluded with 27. With the hierarchical decoder, Full FT was 0.24 points higher ([0.02, 0.47]). Externally, Full FT was 0.9 points lower on CDEMRIS ([−2.4, 0.1]) and 4.5 points higher on bSSFP data ([−0.02, 9.4]). ViT-Large only.
If fully fine-tuning, clip gradients.	Clipping prevented validation collapse in all seeds and raised Large Full FT by 0.59 points; under the common recipe Full FT gave no accuracy advantage over LoRA.
Read features from several encoder blocks, especially for a frozen encoder.	+2.24 (Frozen) and +0.73 (LoRA) points with no extra trainable parameters; with the full common recipe the frozen encoder reached 0.903, 0.83 points below Full FT, at 1.7 GB training memory, but 2.21 points below LoRA with 11 training volumes. A hierarchical decoder with skip connections added at most 0.5 points more and lowered Dice with 11 volumes. Tested on ViT-Large only.
A larger backbone did not pay off in this protocol.	Giant − Large: 0.07 to 0.59 points, 0.15 for LoRA at the higher adapter rate; 2.6 to 3.3 times the training memory of the matched Large configuration.
Do not equate few trainable parameters with low compute.	Giant LoRA: about 135 times fewer trainable parameters than Large Full FT but more memory, time, and GPU-hours.
Report the training recipe together with strategy rankings.	The common improved recipe changed the Full FT−Frozen gap from 3.19 to 0.83 points and the Full FT−LoRA gap from 0.74 to 0.00; single-factor changes moved the Full FT–Giant LoRA gap between 0.13 and 0.72 points.
Where the highest in-domain Dice is the goal, prefer a task-specific nnU-Net, especially with few annotated volumes.	Pipeline-level comparison (3D, augmentation, own schedule; the DINOv2 models used no augmentation): +2.39 points over Large Full FT internally (1.90 over the common-recipe Full FT and LoRA, 1.57 over Full FT with the hierarchical decoder), widening to 3.71 over LoRA with 11 training volumes; inconclusive on external LGE data; higher on bSSFP data.

the annotation actually used exceeds the training count; and nnU-Net kept plans derived from all 123 development volumes.

The conclusions are further restricted to one split, three seeds (one Large LoRA run and two runs of the second from-scratch recipe reached the 50-epoch limit), a 2D formulation without inter-slice modeling, no augmentation, LoRA rank 4, a four-block partial depth, and a decoder whose input width differs between backbones. The nnU-Net comparison is contextual because its recipe differs in many respects. Surface distances are exploratory, and no claims are made about scar delineation, clinical utility, or patient outcomes.

7. Conclusion

AdaDINO-LA characterizes DINOv2 adaptation for LA segmentation from LGE-MRI under one specified protocol. DINOv2 initialization was the largest factor, raising Dice by 14 points over the same ViT trained from scratch. Under the main protocol, Large Full FT gave the highest DINOv2 overlap; under one improved recipe applied post hoc to all ViT-Large strategies, LoRA reached the same mean Dice with about 56% of the training memory and the frozen encoder came within 0.83 points. With 11 instead of 108 training volumes, the benefit of pretraining over an unaugmented ViT trained from scratch grew to 42 points, and LoRA and full fine-tuning still did not differ clearly. A larger encoder added little at a high cost, and a hierarchical decoder with skip connections added at most 0.5 points with all training volumes and lowered Dice with 11. The adapter learning rate, gradient clipping, and especially the decoder readout changed the gaps between strategies, so such comparisons should be reported together with their recipe. A task-specific 3D nnU-Net, the only pipeline trained with augmentation, remained more accurate internally, by a wider margin with fewer training volumes, and on bSSFP data, and the external LGE comparison was inconclusive. Patient independence remains unverified, so internal estimates may be optimistic.

CRediT authorship contribution statement

Quoc Khanh Be: Methodology, Software, Validation, Formal analysis, Investigation, Visualization, Writing – original draft. **Nam Hai Tran:** Investigation, Writing – review & editing. **Gia Duc Trung Nguyen:** Investigation, Writing – review & editing. **Quynh Dao Thi Thuy:** Conceptualization, Supervision, Project administration, Resources, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

Posts and Telecommunications Institute of Technology (PTIT), Hanoi, Vietnam.

Acknowledgements

This work was supported by the Data science Laboratory, Posts and Telecommunications Institute of Technology, Hanoi, Vietnam. The authors would like to thank PTIT for the financial support.

Ethics statement

This study used only publicly available, de-identified imaging data (the 2018 Left Atrial Segmentation Challenge collection, CDEMRIS, and the Medical Segmentation Decathlon) under the providers' terms of use. Ethical approval and informed consent were obtained by the original data providers; no additional approval was required for this secondary analysis.

Data availability

The Utah2018 LGE-MRI data are available from the original challenge provider under its access conditions and are not redistributed by the authors. Source code, configuration files, case lists, per-case metrics, and the scripts that regenerate the reported tables and quantitative figures are publicly available at `https://github.com/qkhanhbe/AdaDINO-LA`. Trained model weights are available from the corresponding author upon reasonable request and will be deposited in a public archive upon publication. Supplementary material reports parameter counts and training lengths (Table S1), exploratory surface metrics (Table S6), validation-trajectory summaries (Table S7), recorded software and source identifiers (Table S8), and validation learning curves (Figures S4–S6).

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used Codex (OpenAI) and Claude (Anthropic) to assist with programming the data-preparation, experiment-management, evaluation, and analysis scripts (Section 4.2) and with drafting and language editing of the manuscript. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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